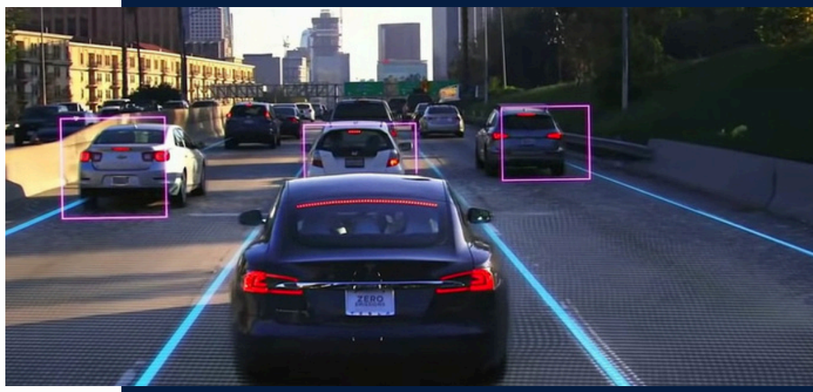
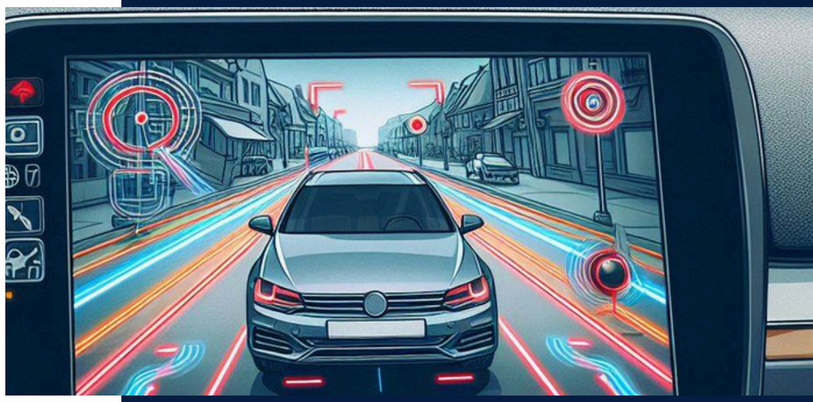
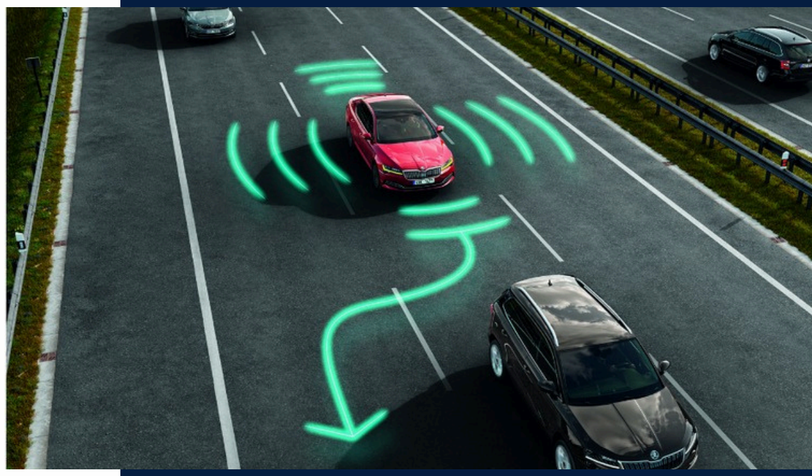
HyperSense







Idea/ solution / prototype

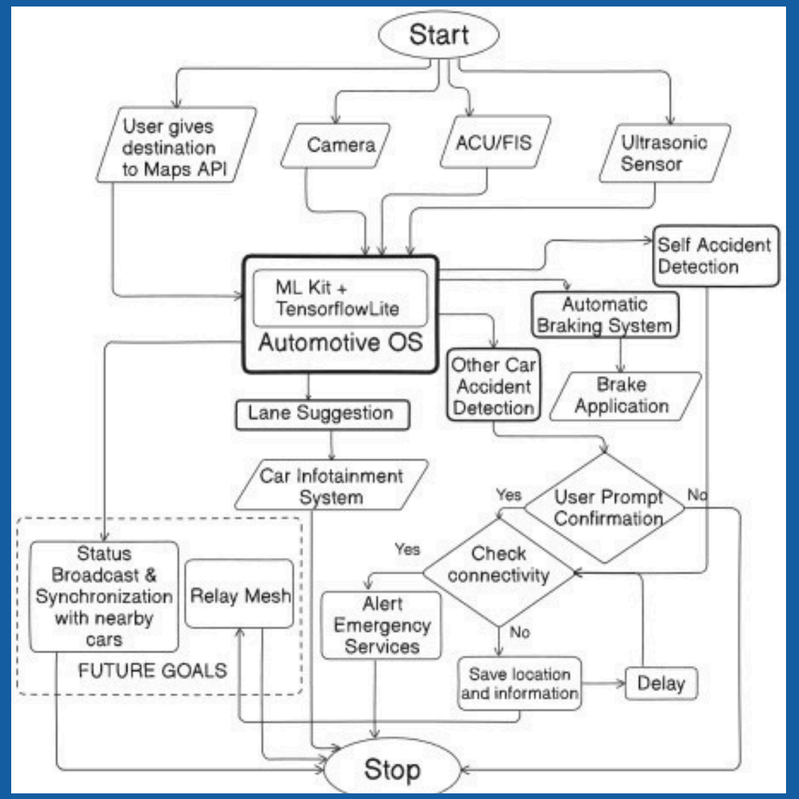
1.Using Google Maps API to set destination and then using front installed camera with real-time object segmentation to provide intelligent lane suggestion.

2.Using ACU & FIS sensor to detect airbag deployment and alert emergency services.

3.Using Ultrasonic sensor to implement Emergency Braking.

4.Using object tracking to detect real-time accidents and object detection to check recent accidents and alert emergency services.

5.Future goals include using FM transmitters to synchronize nearby cars, providing multitude of facilities like multi-car lane suggestion, pothole warning, status broadcast (PANPAN, MAYDAY, NORMAL). Bluetooth 5 serves as backup to FM.











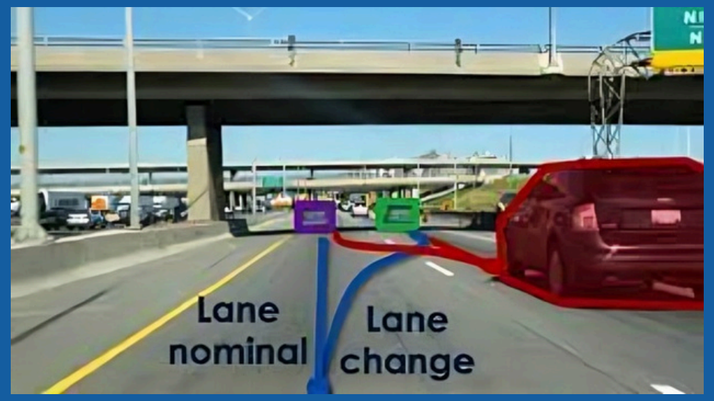


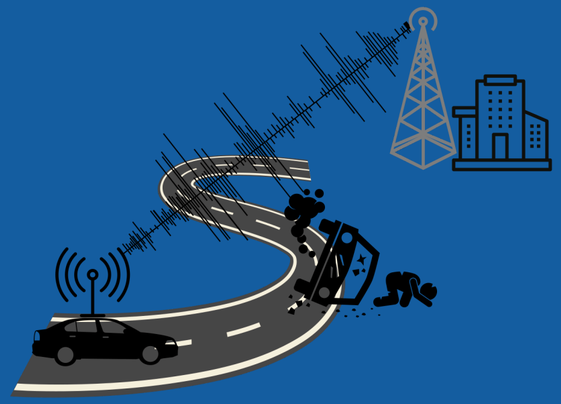














Use Cases

Enhancing Daily Commutes
Deploying live lane suggestions to ease traffic,improve flow, and shorten travel times

Deploying Advanced Collision Units (ACU) and Front Impact Sensors (FIS) for swift accident assistance
Improving Driving Safety Measures

Using passing vehicles to relay SOS messages for emergency responses in remote areas.

Future Innovations: Enhancing driving safety with FM transmitters and Bluetooth 5 for advanced features

Dependencies

1. Google AutomotiveOS
2. Google Play Services
3. Internet Connectivity (Optional)
4. Ultrasonic, ACU & FIS Sensor

Show Stoppers

1. Integration ofAutomotive OS in most cars
2. Hardware processing limitations of cars
3. Reduction of Realtime Latency
4. Abuse prevention of FM frequencies

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Team member 2	mohd bilal	CSE(AIML)	III
Team Member 3	Siddharth	CSE(AIML)	III
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OUR TEAM

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Team Member 7	Janvi	CSE(AIML)	III
Team Member 8	Kavya	Ee	III

OUR TEAM

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Team member 10	Maithili	CSE	III
Team Member 7	Janvi	CSE(AIML)	III
Team Member 8	Kavya	ISE	III